

Movie Industry Analysis: Correlation and Trends

1. Project Overview

This project analyzes the movie industry over a period of four decades (1980–2016) using a dataset of 6,820 movies. The primary goal is to explore the relationship between various factors—such as budget, company, and genre—and a movie's gross earnings to determine which features are the strongest predictors of financial success.

2. Dataset Summary

- **Rows:** 6,820
- **Columns:** 15
- **Key Features:**
 - **Financial Metrics:** Budget and Gross earnings.
 - **Content Attributes:** Movie name, genre, rating, score, and runtime.
 - **Talent and Production:** Director, writer, star, and production company.
 - **Release Information:** Release date, country, and year.
- **Missing Data:** The dataset contains some null values (e.g., movies with 0.0 budget or gross in certain entries) that require cleaning before analysis.

3. Exploratory Data Analysis using Python

The analysis was conducted using Python with libraries such as pandas, numpy, seaborn, and matplotlib.

- **Data Loading:** Imported the movies.csv file into a DataFrame.
- **Data Cleaning:**
 - Checked for missing data using a loop to find the percentage of null values in each column.
 - Removed rows with null values to ensure data integrity.
- **Data Type Correction:** Converted budget and gross columns from floats to integers to

remove trailing zeros and improve readability.

- **Standardization:**
 - Fixed the release year by extracting the year directly from the released column to ensure consistency with the year column.
 - Sorted the data by gross revenue in descending order to identify top earners
- **Feature Engineering:** Created a "Correct Year" column to align with the actual release date.
- **Duplicate Removal:** Ran checks to drop any duplicate entries across the dataset.

4. Data Analysis & Correlation Insights

We performed statistical analysis to identify what drives movie revenue:

1. **Budget vs. Gross Correlation:** Built a scatter plot using regplot which revealed a positive linear relationship between movie budget and gross earnings.
2. **Correlation Matrix:** Generated a heatmap using the Pearson correlation method to visualize relationships between all numeric variables.
3. **Top Correlates:** Identified that **Budget** and **Votes** have the highest correlation with **Gross** revenue.
4. **Genre Performance:** Analyzed revenue distribution across different genres (Action, Adventure, Comedy, etc.).
5. **Company Impact:** Used "Factorizing" techniques to assign numeric codes to non-numeric columns like company, allowing for a comprehensive correlation matrix of all features.
6. **Director & Writer Influence:** Evaluated the success of films based on the involved creative talent.
7. **Runtime Analysis:** Examined if the length of a movie had any significant correlation with its score or revenue.
8. **Score vs. Popularity:** Compared IMDB scores with the number of user votes to see if high-rated movies always generate high engagement.
9. **Production Company Dominance:** Identified the most prolific and high-earning

companies in the dataset.

10. **Yearly Trends:** Tracked how budgets and gross earnings have evolved from 1980 through 2016.

5. Data Visualizations

Visual insights were generated to present findings:

- **Scatter Plots:** Visualizing the trend line between budget and revenue.
- **Correlation Heatmaps:** A color-coded matrix highlighting the strongest and weakest predictors of success.
- **Company Revenue Charts:** Analysis of top-performing production studios.

6. Business Recommendations

- **Strategic Budgeting:** Higher budgets are strongly tied to higher gross earnings; however, this must be balanced with the number of user votes (popularity) to ensure a hit.
- **Audience Engagement:** Focus on marketing strategies that drive user engagement (votes), as high vote counts are strongly correlated with financial success.
- **Genre Selection:** Invest in genres that historically show higher gross potential based on the dataset trends.
- **Talent Retention:** Partner with directors and stars associated with high-grossing films, as creative talent is a key component of the movie's marketability.